MPHIL CLINICAL PSYCHOLOGY • RESEARCH SYNOPSIS • 2025-2027

MAN College of Special Education & Psychological Studies, Guna (MP)

Efficacy of Brief Mindfulness-Based Relapse Prevention (B-MBRP) Intervention on Craving, Impulsivity, and Mindfulness in Substance-Dependent Patients

A Pre-Test Post-Test Control Group Experimental Design

Researcher: Tejas Dangodra | **Guide:** Dr. Himanshi Singh

Department of Clinical Psychology

Krantivir Tatya Tope Vishwavidyalaya, Guna, Madhya Pradesh

Approved by Rehabilitation Council of India (RCI)

01 Introduction: Substance Dependence in India

INDIAN CONTEXT

MAGNITUDE study (2019): ~3.1 crore individuals affected by SUDs. Opioid dependence is major burden in Punjab, Rajasthan, NE India. WHO estimates India accounts for ~25% of global opioid-related deaths in SE Asia.

TREATMENT GAP

Indian de-addiction centres (including the present study setting) primarily offer pharmacotherapy (OST, naltrexone) with limited structured psychotherapy. Psychosocial interventions remain under-utilized despite evidence of superior combined outcomes.

RELAPSE CRISIS

Relapse rates: 40-60% within first year globally (NIDA, 2020). Indian opioid studies report **70-80%** (Mattoo et al., 2009). Triggers: craving, negative affect, interpersonal conflict, environmental cues.

WHY BRIEF INTERVENTIONS?

Indian rehab settings have limited resources & short admission windows (4-6 weeks). Traditional 8-week MBRP is impractical. Brief 6-session adaptation is essential for feasibility at district-level facilities.

KEY ABBREVIATIONS: MBRP = Mindfulness-Based Relapse Prevention | TAU = Treatment As Usual | OCDUS = Obsessive Compulsive Drug Use Scale | BIS-11 = Barratt Impulsiveness Scale | FFMQ = Five Facet Mindfulness Questionnaire | ASSIST = WHO Substance Screening Test

02 Introduction: What is MBRP?

Mindfulness-Based Relapse Prevention (MBRP) was developed by Bowen, Chawla & Marlatt (2011) at the University of Washington. It integrates mindfulness meditation practices with cognitive-behavioral Relapse Prevention framework.

CORE MECHANISM

Teaches patients to **observe craving and emotional distress as transient mental events** without automatically acting on them. Key technique: *"Urge Surfing"* — riding the wave of craving until it passes naturally.

NEUROBIOLOGICAL BASIS

Mindfulness strengthens **prefrontal cortex regulation** over amygdala reactivity. Enhances PFC-limbic connectivity, improves inhibitory control, and reduces automatic craving-use pathways.

Brief B-MBRP Adaptation: 6 structured sessions over 3 weeks | Group format (6-8 patients) | 45 min/session | Twice weekly | Feasible within typical Indian IPD admission window of 4-6 weeks.

TARGETS CRAVING

Urge surfing disrupts automatic craving-use cycle

TARGETS IMPULSIVITY

Mindful pause strengthens response inhibition

BUILDS MINDFULNESS

Structured meditation cultivates dispositional awareness

03 Variable 1: Craving

DEFINITION

Clinical: Intense, overwhelming desire to use a substance (DSM-5 & ICD-11 core feature).
Theoretical: Obsessive thoughts + compulsive urges related to drug use (Franken et al., 2002)

RELEVANCE IN SUBSTANCE DEPENDENCE

Primary trigger for relapse. Craving intensity predicts treatment dropout, lapse episodes, and full relapse (Tiffany & Wray, 2012). Traditional suppression paradoxically increases intensity through rebound effects.

NEUROPSYCHOLOGICAL BASIS

Mesolimbic dopamine pathway activation. Ventral striatum & OFC hyperactivity to drug cues. Incentive sensitization (Robinson & Berridge, 1993). PFC hypoactivation during craving episodes.

MBRP TECHNIQUES TARGETING CRAVING

- **Urge Surfing:** Observe craving as transient wave
- **SOBER Space:** Stop-Observe-Breathe-Expand-Respond
- **Cognitive Decentering:** "I am having a craving"
- **Mindfulness Exposure:** Non-reactive awareness

04 Variable 2: Impulsivity

DEFINITION

Clinical: Rapid, unplanned actions without considering consequences (Moeller et al., 2001).
Theoretical: Three dimensions — Motor + Attentional + Non-Planning Impulsivity (Patton et al., 1995)

RELEVANCE IN SUBSTANCE DEPENDENCE

- Higher trait impulsivity predicts relapse
- Mediates craving-to-use behavior
- Associated with treatment non-adherence & dropout
- Both a risk factor for and consequence of chronic use

NEUROPSYCHOLOGICAL BASIS

PFC dysfunction impaired executive control. Reduced inhibitory control (Go/No-Go paradigms). Impaired delay discounting. DLPFC hypoactivation in substance users.

MBRP TECHNIQUES TARGETING IMPULSIVITY

- **Response Inhibition:** Mindful pause before action
- **STOP Technique:** Stop-Take a breath-Observe-Proceed
- **Awareness Training:** Notice impulse-action sequences
- **Mindful Decision-Making:** Space between stimulus & response

05 Variable 3: Mindfulness

DEFINITION

Clinical: Present-moment awareness with openness & non-judgment.
Theoretical: 5 facets — Observing, Describing, Acting with Awareness, Non-Judging, Non-Reactivity (Baer et al., 2006)

RELEVANCE IN SUBSTANCE DEPENDENCE

Substance users show significantly lower dispositional mindfulness. Acts as protective factor against relapse triggers. Improvements mediate MBRP treatment outcomes across studies.

NEUROPSYCHOLOGICAL BASIS

ACC activation enhances self-regulation. Insula mediates interoceptive awareness. PFC-amygdala connectivity improves emotional regulation. DMN regulation reduces rumination.

MBRP TECHNIQUES FOR BUILDING MINDFULNESS

- **Sitting Meditation:** Focused attention on breath
- **Body Scan:** Non-judgmental bodily awareness
- **Mindful Movement:** Present-moment yoga/walking
- **Non-Judgmental Awareness:** Labeling without evaluation

06 Review of Literature (1/5): MBRP & Relapse Prevention

BOWEN ET AL. (2014) — JAMA PSYCHIATRY

N=286 RCT. At 12-month follow-up, MBRP participants reported significantly **fewer days of substance use** and heavy drinking compared to standard RP and TAU. MBRP maintained superior long-term outcomes through cultivation of mindfulness skills as durable protective mechanism.

BOWEN & MARLATT (2009) — PSYCH. OF ADDICTIVE BEHAVIORS

Brief urge surfing meditation with incarcerated substance users. **Significant reductions in craving intensity and frequency** vs. controls. Validates that even brief mindfulness exposure can disrupt automaticity of craving responses. Supports feasibility of 6-session B-MBRP protocols.

Key Insight: Both studies establish that MBRP integrates present-moment awareness with cognitive-behavioral strategies, creating synergistic therapeutic effects addressing both automatic reactivity and cognitive distortions that precipitate relapse.

07 Review of Literature (2/5): Craving & Mindfulness Mechanisms

GARLAND ET AL. (2014) — FRONTIERS IN PSYCHIATRY

Demonstrates MORE reduces opioid craving through: (1) attentional reorientation away from drug cues, (2) positive reappraisal of neutral stimuli, (3) enhanced savoring of healthy pleasures. **Neuroimaging shows mindfulness modulates prefrontal and limbic craving circuits.**

WITKIEWITZ ET AL. (2013) — ADDICTIVE BEHAVIORS

Secondary RCT analysis. Over 4-month follow-up, MBRP participants showed lower craving AND the **affect-craving pathway was significantly attenuated.** Mindfulness weakens the link between negative emotions and automatic craving by cultivating non-reactive awareness.

Key Mechanism: MBRP decouples the affect-craving pathway. Negative emotions no longer automatically trigger craving responses — this distinguishes MBRP from traditional RP approaches.

08 Review of Literature (3/5): Impulsivity & Mindfulness

GARLAND ET AL. (2016) — J CONSULTING & CLINICAL PSYCHOLOGY

RCT with substance-dependent adults. Mindfulness-Oriented Recovery Enhancement produced **significant BIS-11 reductions in motor and attentional impulsivity.** Mechanism: mindfulness strengthens prefrontal cortical inhibitory control through sustained attention and deliberate non-reactivity training.

MURPHY & MACKILLOP (2012) — PSYCHOPHARMACOLOGY

N=340 cross-sectional. Trait mindfulness **inversely associated with impulsive decision-making** (delay discounting). Mindfulness moderated the impulsivity-substance use link. Higher mindfulness = weaker association between impulsivity and alcohol consequences.

Implication: Mindfulness functions as a cognitive resource enabling impulsive individuals to override automatic behavioral tendencies through enhanced metacognitive awareness and response flexibility.

09 Review of Literature (4/5): Meta-Analyses & Brief Models

LI ET AL. (2017) — J SUBSTANCE ABUSE TREATMENT

Meta-analysis of **42 RCTs.** Effect sizes: substance misuse (d=0.33), craving (**d=0.68**), stress (d=0.44). **Brief interventions (4-8 sessions) showed comparable efficacy** to longer protocols when appropriately structured. Supports present study's 6-session design.

GLASNER-EDWARDS ET AL. (2017) — MINDFULNESS

Pilot RCT: Abbreviated **6-session MBRP** for stimulant-dependent adults. Both feasible and effective in reducing substance use frequency and craving intensity vs. health education control. **Directly validates the brief intervention model** proposed in present research.

Critical Evidence: 6-session MBRP protocols can be successfully implemented without substantial loss of efficacy, making them appropriate for Indian settings with 4-6 week admission durations.

10 Review of Literature (5/5): Indian Context

GHOSH ET AL. (2018) — INDIAN J PSYCHIATRY

Relapse rates **exceed 70%** among opioid-dependent patients in North Indian de-addiction centres within 3 months post-discharge. Primary determinants: craving, peer influence, negative affect, lack of psychological aftercare. Indian facilities predominantly rely on pharmacology.

SARKAR & BALHARA (2016) + JAIN ET AL. (2013)

Sarkar: Highlights underutilization of structured psychological interventions in Indian de-addiction. Barriers: limited trained personnel, absent validated protocols.

Jain: Preliminary evidence from MBI with alcohol-dependent patients in India showing initial craving reductions.

CRITICAL GAP: Indian treatment facilities predominantly rely on pharmacological approaches with minimal integration of evidence-based psychological interventions specifically designed for relapse prevention in opioid-dependent populations.

11 Research Gap

1. **Limited evidence for brief MBRP adaptations in LMIC settings:** The majority of MBRP research evaluates the standard 8-week protocol in Western outpatient settings. Condensed protocols suitable for resource-constrained inpatient environments remain insufficiently examined (Li et al., 2017)
2. **Absence of simultaneous multi-domain outcome assessment:** Existing studies typically examine craving, impulsivity, or mindfulness in isolation. The interplay between these three mechanistically linked variables within a single intervention framework has not been adequately investigated in Indian clinical populations
3. **Paucity of culturally contextualized MBRP evidence from India:** Despite India's substantial opioid dependence burden, empirical validation of MBRP-based protocols in Indian de-addiction infrastructure remains virtually absent from the published literature
4. **Insufficient integration of psychological interventions in Indian de-addiction practice:** Current treatment models at district-level centers remain predominantly pharmacological, with limited evidence-based psychosocial adjuncts (Sarkar & Balhara, 2016; Ghosh et al., 2018)
5. **Need for attention-matched controlled designs:** Most existing mindfulness studies in Indian addiction settings lack active control conditions, making it difficult to distinguish mindfulness-specific effects from non-specific therapeutic factors

PRESENT STUDY ADDRESSES: First brief MBRP trial (6 sessions / 3 weeks) in Indian opioid-dependent sample at a de-addiction centre with active control and simultaneous three-variable assessment.

12 Aim of the Study

To evaluate the efficacy of a **Brief Mindfulness-Based Relapse Prevention (B-MBRP)** intervention (6 sessions over 3 weeks) in reducing craving and impulsivity, and enhancing mindfulness, among male substance-dependent patients at a de-addiction centre.

Comparing: Brief MBRP + TAU (Experimental) vs. Psychoeducation + TAU (Active Control)

60	6	3	3
Total Participants	B-MBRP Sessions	Weeks Duration	Outcome Variables

13 Objectives

1. To assess and compare **craving levels** (pre vs. post) in Experimental (Brief MBRP + TAU) and Control (Psychoeducation + TAU) groups
2. To assess and compare **impulsivity levels** (pre vs. post) in both groups
3. To assess and compare **mindfulness levels** (pre vs. post) in both groups
4. To determine whether Brief MBRP + TAU is significantly more effective than Psychoeducation + TAU in **reducing craving**
5. To determine whether Brief MBRP + TAU is significantly more effective than Psychoeducation + TAU in **reducing impulsivity**
6. To determine whether Brief MBRP + TAU is significantly more effective than Psychoeducation + TAU in **enhancing mindfulness**

14 Hypotheses (Null Hypotheses Only)

$H_0$1 — CRAVING

There is no significant difference in craving scores (OCDUS) between the Experimental Group (Brief MBRP + TAU) and the Control Group (Psychoeducation + TAU) from pre-test to post-test.

$H_0$2 — IMPULSIVITY

There is no significant difference in impulsivity scores (BIS-11) between the Experimental Group (Brief MBRP + TAU) and the Control Group (Psychoeducation + TAU) from pre-test to post-test.

$H_0$3 — MINDFULNESS

There is no significant difference in mindfulness scores (FFMQ) between the Experimental Group (Brief MBRP + TAU) and the Control Group (Psychoeducation + TAU) from pre-test to post-test.

Statistical Testing: Alpha = 0.05 (two-tailed) | Primary Analysis: ANCOVA with pre-test scores as covariates.

15 Operational Definitions of Key Terms (1/2)

Term	Operational Definition
Craving	"Craving" in this study refers to the intensity and frequency of obsessive thoughts about drug use and compulsive urges to use opioids, as measured by the total score on the **Obsessive Compulsive Drug Use Scale (OCDUS;** Franken et al., 2002). Scores range from 0 to 52; higher scores indicate greater craving.
Impulsivity	"Impulsivity" in this study refers to the multidimensional tendency to act without adequate forethought, encompassing motor, attentional, and non-planning components, as measured by the total score and three subscale scores on the **Barratt Impulsiveness Scale-11 (BIS-11;** Patton et al., 1995). Scores range from 30 to 120; higher scores indicate greater impulsivity.
Mindfulness	"Mindfulness" in this study refers to the dispositional capacity for present-moment awareness with non-judgment and non-reactivity, as measured by the total score on the **Five Facet Mindfulness Questionnaire (FFMQ;** Baer et al., 2006) across five facets. Scores range from 39 to 195; higher scores indicate greater mindfulness.

16 Operational Definitions of Key Terms (2/2)

Term	Operational Definition
Brief MBRP	"Brief MBRP" in this study refers to a structured 6-session mindfulness-based relapse prevention intervention delivered twice weekly over 3 weeks (45 min/session) in group format (6-8 patients), adapted from Bowen, Chawla & Marlatt (2011), incorporating body scan, breath meditation, urge surfing, SOBER breathing space, and relapse prevention planning.
Substance Dependence	"Substance Dependence" in this study refers to a clinical diagnosis of Substance Dependence Syndrome as per ICD-10 (F10-F19) criteria, with primary opioid dependence (heroin or pharmaceutical opioids), confirmed by a qualified psychiatrist at a de-addiction centre.
Treatment As Usual (TAU)	"TAU" in this study refers to the standard pharmacological treatment at the de-addiction centre, including medically supervised detoxification, OST (buprenorphine/methadone), naltrexone maintenance, routine counseling, and daily ward activities.
Psychoeducation	"Psychoeducation" in this study refers to 6 structured informational sessions (45 min, twice weekly, 3 weeks) covering addiction science, effects of opioids, relapse warning signs, health/nutrition, social consequences, and recovery motivation — matched for time and attention but containing NO mindfulness component.

17 Research Design: Pre-Test Post-Test Control Group

Design Notation:

R O_1 X_1 O_2 ▯ Experimental Group (Brief MBRP + TAU)

R O_1 X_2 O_2 ▯ Control Group (Psychoeducation + TAU)

RANDOMIZE (N=60) ▯ **PRE-TEST (O_1) OCDUS+BIS-11+FFMQ+ASSIST** ▯ **INTERVENTION 3 Weeks** ▯ **POST-TEST (O_2) OCDUS+BIS-11+FFMQ**

DESIGN FEATURES

True experimental | Random assignment | Active control (attention-matched) | TAU continued for all | Pre-post measurement at standardized time points

SETTING

De-addiction Centre, Guna, MP | Assessments by blinded RA | Intervention by MPhil Clinical Psychologist (Researcher)

18 Sample & Sampling Strategy

STAGE 1: PURPOSIVE SELECTION

Eligible participants identified based on inclusion/exclusion criteria from patients admitted to the de-addiction centre. Consecutive sampling of all eligible patients during recruitment window.

STAGE 2: RANDOM ASSIGNMENT

Computer-generated randomization allocating eligible participants to Experimental (n=30) or Control (n=30) groups using sealed opaque envelopes.

60	70	Male	18-50
Total (30/group)	Recruit (attrition)	Gender	Age Range

Male-Only Justification: Indian de-addiction centres admit ~90-95% males. MAGNITUDE study reports male:female ratio ~10:1 for opioid dependence. Homogeneous sample strengthens internal validity. Female-specific studies recommended as future direction.

19 Sample Size Estimation

$$n = [(Z\alpha/2 + Z\beta)^2 \times 2 \times \sigma^2] / d^2$$

PARAMETERS

- Effect size (d) = 0.50 (medium)
- Power ($1-\beta$) = 0.80 □ $Z\beta = 0.84$
- Alpha = 0.05 (two-tailed) □ $Z\alpha/2 = 1.96$

CALCULATION

$n = [(1.96 + 0.84)^2 \times 2 \times 1] / (0.50)^2$

$n = [7.84 \times 2] / 0.25 = 62.72$

n □ 63 total (~32 per group)

FINAL DECISION: N = 60 (30 per group). ANCOVA reduces required n. G*Power 3.1 verification confirms adequacy. Recruit 70 total (35/group) for ~15% attrition. Consistent with Glasner-Edwards et al. (2017) and Bowen & Marlatt (2009).

20 Inclusion Criteria

1. Diagnosis of **Substance Dependence** as per ICD-10 (F10-F19) / ICD-11 criteria
2. Primary substance: **Opioids** (heroin, pharmaceutical opioids); polysubstance with primary opioid included
3. **Male** participants aged **18-50 years**
4. Completed **detoxification phase** (minimum 7 days post-withdrawal)
5. Currently admitted at the **de-addiction centre**
6. Minimum education: **5th standard** (ability to comprehend psychometric tools)
7. Willingness to provide **written informed consent**
8. Able to attend all **6 intervention sessions** during 3-week period

21 Exclusion Criteria

1. **Severe psychiatric comorbidity:** Psychotic disorders, Bipolar I, severe MDE with suicidality
2. **Significant cognitive impairment** (MMSE < 24) or intellectual disability
3. **Active withdrawal symptoms** (COWS score > 12)
4. History of **traumatic brain injury** with LOC > 30 minutes
5. Current participation in **another structured psychological intervention** study
6. **Medical instability** requiring acute/intensive care
7. History of prior formal **mindfulness/meditation training** exceeding 1 month

22 Variables of the Study

Type	Variable	Levels / Measurement
Independent Variable	Type of Intervention	Level 1: Brief MBRP + TAU (Exp) Level 2: Psychoeducation + TAU (Control)
DV 1	Craving	OCDUS total score (0-52)
DV 2	Impulsivity	BIS-11 total + 3 subscales (30-120)
DV 3	Mindfulness	FFMQ total + 5 facets (39-195)

Controlled Variables: Age, education, duration of use, severity (ASSIST baseline), TAU components constant across groups, session duration equalized (45 min × 6 sessions for both groups)

23 Tool 1: OCDUS (Craving Measure)

Obsessive Compulsive Drug Use Scale (OCDUS) — Franken et al. (2002)

13-item self-report measuring obsessive thoughts about drug use and compulsive urges. Captures both cognitive (obsessive) and behavioral (compulsive) craving dimensions.

SCORING

5-point scale (0-4) | Total range: 0-52 | Higher = greater craving
Subscales: (1) Obsessive thoughts/interference, (2) Desire/control, (3) Resistance to thoughts

PSYCHOMETRIC PROPERTIES

Internal consistency: α = 0.86-0.90
Test-retest: r = 0.78
Convergent with VAS craving: r = 0.55-0.67
Predicts relapse; sensitive to treatment changes

Justification: Applicable across substances including opioids. Brief (5 min). Captures multidimensional craving. Suitable for pre-post. Adaptable for Hindi administration.

24 Tool 2: BIS-11 & Tool 3: FFMQ

BARRATT IMPULSIVENESS SCALE (BIS-11) — PATTON ET AL., 1995

- 30 items | 4-point scale (1-4) | Range: 30-120
- 3 factors: Attentional, Motor, Non-Planning
- Higher = greater impulsivity
- α = 0.79-0.83; test-retest r = 0.83
- Discriminates SUD from controls
- **Hindi validated version available**

FIVE FACET MINDFULNESS QUESTIONNAIRE (FFMQ) — BAER ET AL., 2006

- 39 items | 5-point scale (1-5) | Range: 39-195
- 5 facets: Observing, Describing, Acting with Awareness, Non-Judging, Non-Reactivity
- Higher = greater mindfulness
- α = 0.75-0.91 across facets
- Sensitive to mindfulness interventions
- **Hindi adaptation available**

Both tools: Self-report | Validated in substance use populations | Hindi available | Administered at PRE-TEST and POST-TEST | 10-15 min each

25 Tool 4: WHO-ASSIST V3.0 (Baseline Severity)

WHO-ASSIST Version 3.0 — Alcohol, Smoking and Substance Involvement Screening Test (WHO ASSIST Working Group, 2002; WHO, 2010)

8-item questionnaire measuring individual's use of alcohol, tobacco, and other drugs across lifetime and past 3 months. Screens across 10 substance categories. **V3.0 is the WHO-recommended standard** for clinical and research screening globally.

SCORING (V3.0)

Substance-specific risk:
Low: 0-3 (no intervention)
Moderate: 4-26 (brief intervention)
High: 27+ (referral to specialist)

PSYCHOMETRICS

Test-retest: r = 0.58-0.90
Alpha: α = 0.77-0.94
Sensitivity: 0.80
Specificity: 0.71
Validated across 18 countries

PURPOSE IN STUDY

PRE-TEST ONLY
Establish baseline severity
Ensure group equivalence
Stratification for randomization
NOT an outcome measure

V3.0 Advantages: WHO-validated internationally | Hindi version available | Used in NDDTC & NIDA studies | Brief (5-10 min) | Linked to brief intervention guidelines | Recommended for primary care & specialist settings

26 Data Collection Procedure: Step-by-Step

Screen Consent Pre-Test Randomize Intervene (3 wks) Post-Test Analyze

1. **Screening (Day 1-3):** Review admission records. Identify male patients (18-50) with opioid dependence (ICD-10), 7+ days detox. Check MMSE 24, COWS 12. Apply inclusion/exclusion criteria.
2. **Informed Consent (Day 3-4):** Individual meeting in Hindi. Explain study purpose, procedures, duration, voluntary nature, right to withdraw, confidentiality. Obtain written signed consent.
3. **Pre-Test Assessment (Day 4-5):** Administer in quiet room: ASSIST (severity baseline) + OCDUS (craving) + BIS-11 (impulsivity) + FFMQ (mindfulness). ~40 min total. Conducted by blinded RA.
4. **Randomization (Day 5):** Computer-generated random sequence. Sealed opaque envelopes. Allocate to Experimental (MBRP) or Control (Psychoeducation). Stratified by ASSIST severity.
5. **Intervention (Weeks 1-3):** 6 sessions, twice weekly, 45 min each. Groups of 6-8. MBRP group: mindfulness techniques. Control: psychoeducation. Both continue TAU throughout.
6. **Post-Test (Within 1 week post-intervention):** Re-administer OCDUS + BIS-11 + FFMQ only (no ASSIST). Same blinded RA. Same conditions as pre-test.

27 Data Collection: Techniques & Conditions

ASSESSMENT SETTING

- Quiet, private room within the de-addiction centre
- Comfortable seating, adequate lighting
- No distractions or other patients present
- Morning hours (9-12 AM) to avoid medication effects
- Same room & conditions for pre & post

ADMINISTRATION PROTOCOL

- Researcher-assisted (read aloud if literacy issues)
- Hindi language throughout
- Standardized instructions read verbatim
- Unlimited time (typically 35-45 min total)
- Practice items completed first to ensure comprehension

BLINDING & BIAS CONTROL

- Assessor (RA) blinded to group allocation
- Participants instructed not to reveal group
- Standardized order: OCDUS □ BIS-11 □ FFMQ
- Social desirability managed via anonymous coding
- No researcher present during assessment

DATA MANAGEMENT

- Participant ID codes (no names on forms)
- Double data entry for accuracy
- Locked cabinet for paper forms
- Electronic data password-protected
- Access restricted to research team only

28 B-MBRP Techniques Used in the Study

Technique	Description	Target Variable
Body Scan Meditation	Systematic attention to bodily sensations from head to toe without judgment	Mindfulness (Observing)
Breath Awareness	Focused attention on natural breathing rhythm as present-moment anchor	Mindfulness (Acting with Awareness)
Urge Surfing	Observing craving as a wave that rises, peaks, and falls without acting	Craving reduction
SOBER Breathing Space	Stop-Observe-Breathe-Expand-Respond (3-min emergency technique)	Impulsivity (response inhibition)
Trigger Mapping	Identifying personal high-risk situations, people, places, emotions	Craving awareness
Cognitive Decentering	Labeling: "I am having the thought that..." Creates distance from automatic patterns	Impulsivity + Craving
Loving-Kindness Meditation	Cultivating self-compassion and positive affect; counters shame/guilt cycle	Mindfulness (Non-Judging)
Mindful Movement	Gentle yoga/walking with full present-moment attention to body	Mindfulness (Observing)

29 B-MBRP Session-by-Session Protocol

Session	Title	Activities (45 min)	Home Practice
S1	Autopilot & Awareness	Welcome (5) □ Raisin exercise (10) □ Discussion: automatic patterns (10) □ Brief body scan (12) □ Assign practice (8)	5 min daily body scan
S2	Triggers & Body Scan	Practice review (5) □ Full body scan (15) □ Trigger mapping exercise (15) □ Body signals of craving discussion (10)	Body scan + trigger diary
S3	Breath & SOBER Space	Review (5) □ Breath meditation (10) □ Teach SOBER (10) □ Role-play high-risk scenario (10) □ Integration (10)	Breath + 3x daily SOBER
S4	Urge Surfing & Decentering	Review (5) □ Guided urge surfing (15) □ Cognitive decentering exercise (10) □ Group sharing (15)	Urge surfing when craving arises
S5	Acceptance & Non-Reactivity	Review (5) □ Open awareness meditation (12) □ Acceptance vs avoidance discussion (10) □ Skillful action planning (10) □ Wrap-up (8)	Open awareness daily
S6	Integration & Maintenance	Review (5) □ Loving-kindness meditation (10) □ Personal relapse prevention plan (15) □ Group feedback & closure (15)	Personalized daily plan

Delivery: Twice weekly | 45 min/session | Group (6-8) | Hindi audio meditations provided | Practice logs maintained | Adapted from Bowen, Chawla & Marlatt (2011)

30 How Each B-MBRP Session is Conducted

Standard 6-Step Session Structure (every session follows this format):

1. **Opening Meditation (5 min):** Brief breath awareness to settle the group. Participants sit comfortably with eyes closed. Therapist guides attention to breath in Hindi.
2. **Practice Review (5 min):** Participants share home practice experiences. Troubleshoot difficulties. Normalize challenges. Non-judgmental discussion.
3. **Core Meditation/Exercise (12-15 min):** Session-specific guided meditation (body scan, urge surfing, etc.). Hindi pre-recorded audio + live guidance by therapist.
4. **Psychoeducation & Discussion (10-12 min):** Brief teaching on session theme. Group discussion linking mindfulness to personal recovery. Culturally relevant examples.
5. **Experiential Exercise (8-10 min):** Active practice or role-play (SOBER in high-risk scenario, trigger mapping on paper, decentering with craving thoughts).
6. **Closing & Home Practice (5 min):** Summary of key learning. Assign specific daily practice (5-10 min). Distribute audio. Record attendance.

Therapist: MPhil Clinical Psychologist trained in MBRP | Manualized protocol ensures fidelity | Supervision provided by guide

31 Control Group: Psychoeducation (6 Sessions / 3 Weeks)

Session	Topic	Content
S1	Understanding Addiction	Disease model of addiction; brain changes with chronic use; genetic & environmental factors
S2	Effects of Opioids	Physical consequences (liver, heart, immune); psychological effects; withdrawal timeline
S3	Understanding Relapse	Warning signs; high-risk situations; Marlatt's relapse model; cognitive distortions
S4	Health & Nutrition	Physical recovery during treatment; sleep hygiene; exercise benefits; nutrition for recovery
S5	Social Consequences	Family impact; legal issues; stigma; workplace; rehabilitation resources
S6	Motivation & Goals	Stages of change; recovery planning; goal setting; community resources; discharge planning

Active Control Design: Matched for time (45 min), format (group 6-8), frequency (2x/week), attention, and therapist contact. Uses handouts + visual aids + discussion. Contains **NO mindfulness component** whatsoever.

32 Data Analysis Plan

STATISTICAL TESTS

- Descriptive: Mean, SD, frequencies
- Normality: Shapiro-Wilk test
- Within-group: Paired t-test / Wilcoxon
- Between-group: Independent t-test / Mann-Whitney
- **Primary: ANCOVA**
- Effect size: Partial η^2, Cohen's d
- Alpha = 0.05 (two-tailed) | SPSS 26.0

ANCOVA MODEL

DV: Post-test scores (OCDUS / BIS-11 / FFMQ)
IV: Group (Experimental vs. Control)
Covariate: Pre-test scores (same measure)
ITT: Last Observation Carried Forward (LOCF)

ANCOVA Justification: (a) Controls for baseline differences, (b) Increases power by reducing error variance, (c) More precise treatment effect estimate, (d) Reduces required sample size, (e) Recommended for pre-post designs (Tabachnick & Fidell, 2013)

33 Ethical Considerations

1. **Informed Consent:** Written consent in Hindi; fully informed of purpose, procedures, risks, benefits
2. **Voluntary Participation:** Right to withdraw anytime without penalty or impact on treatment
3. **Confidentiality:** Data coded with participant IDs; no identifying info in publications; secure locked storage
4. **Non-Maleficence:** Control receives active psychoeducation (not waitlist); TAU continued for all
5. **Institutional Approval:** Ethical clearance from IEC prior to data collection
6. **Debriefing:** Control group offered brief MBRP orientation post-study
7. **Compliance:** ICMR (2017) National Ethical Guidelines for Biomedical and Health Research

34 Expected Results

CRAVING (OCDUS)

Significant **REDUCTION** in MBRP group vs. Control

Expected: d = 0.50-0.80

Mechanism: Urge surfing disrupts automatic craving-use cycle

IMPULSIVITY (BIS-11)

Significant **REDUCTION** (Motor + Attentional subscales)

Expected: d = 0.40-0.60

Mechanism: Mindfulness strengthens PFC inhibitory control

MINDFULNESS (FFMQ)

Significant **INCREASE** (Acting with Awareness & Non-Reactivity facets)

Expected: d = 0.50-0.70

Mechanism: Structured meditation builds dispositional mindfulness

Overall: Brief MBRP + TAU expected to demonstrate superiority across all 3 DVs. Null hypotheses expected to be rejected. Supports feasibility of 6-session B-MBRP at a de-addiction centre.

35 Clinical Implications

1. **Validates a brief MBRP model** (6 sessions / 3 weeks) feasible for Indian de-addiction settings with limited resources
2. **Evidence-based psychological intervention** to complement pharmacotherapy (OST, naltrexone) in routine care
3. **Cultural compatibility:** Demonstrates mindfulness approaches are appropriate for Indian populations
4. **Multi-target protocol:** Addresses craving + impulsivity + mindfulness simultaneously through single integrated intervention
5. **Task-shifting potential:** Brief MBRP deliverable by MPhil-trained Clinical Psychologists in district-level settings
6. **Scalable model:** If effective, can be disseminated to government de-addiction centres across India

36 Limitations

1. **Male-only sample** from single center limits generalizability to females and other settings
2. **Short-term assessment:** Post-test immediately after intervention; no long-term follow-up data
3. **Self-report measures** (OCDUS, BIS-11, FFMQ) susceptible to social desirability bias
4. **No biological markers:** Craving measured subjectively without physiological corroboration
5. **Therapist effects:** Single therapist delivery may introduce confounds (mitigated by manualized protocol)
6. **Attention-matched control** does not fully isolate mindfulness-specific mechanisms
7. **Potential attrition** despite over-recruitment (common in substance-dependent populations)

37 Future Directions

FOLLOW-UP & REPLICATION

- 3-month and 6-month follow-up
- Multi-site RCTs across India
- Include female participants

MECHANISM RESEARCH

- Neuroimaging (fMRI/EEG)
- Mediator analysis: mindfulness as mediator
- Dose-response: 4 vs 6 vs 8 sessions

COMPARATIVE STUDIES

- MBRP vs. CBT vs. ACT in India
- Technology-assisted app-based MBRP
- Post-discharge booster sessions

DISSEMINATION

- Develop Hindi MBRP training manual
- Train counselors in govt centers
- Rural access via telehealth delivery

38 Study Summary

Component	Details
Design	Pre-test Post-test Control Group Experimental Design (True Experimental)
Setting	De-addiction Centre, Guna, Madhya Pradesh
Population	Male opioid-dependent patients, aged 18-50, N = 60 (30/group)
Experimental	Brief MBRP (6 sessions / 3 weeks) + Treatment As Usual
Control	Psychoeducation (6 sessions / 3 weeks) + Treatment As Usual
DVs	Craving (OCDUS) + Impulsivity (BIS-11) + Mindfulness (FFMQ)
Sampling	Two-stage: Purposive selection Computer-generated random assignment
Primary Analysis	ANCOVA controlling for pre-test scores as covariates
Hypotheses	Null: No significant difference between groups on OCDUS, BIS-11, FFMQ

SIGNIFICANCE: First brief MBRP trial in Indian opioid-dependent sample addressing critical research gap in evidence-based psychological interventions for de-addiction treatment.

39 Conclusion

1. Substance dependence (particularly opioid) represents a significant public health crisis in India with relapse rates exceeding 70%
2. Current Indian de-addiction treatment relies heavily on pharmacotherapy with limited evidence-based psychological interventions
3. MBRP offers a theoretically grounded approach targeting craving (urge surfing), impulsivity (mindful pause), and mindfulness enhancement simultaneously
4. A brief 6-session B-MBRP protocol is clinically practical, culturally appropriate, and feasible at a de-addiction centre
5. If supported, Brief MBRP can be integrated into standard de-addiction protocols across Indian government rehabilitation centers

> **This study represents a critical first step** in establishing an evidence base for brief mindfulness-based psychological interventions within the Indian de-addiction treatment infrastructure.

References (1/4)

• Baer, R. A., et al. (2006). Using self-report methods to explore facets of mindfulness. *Assessment*, 13(1), 27-45.
• Bowen, S., Chawla, N., & Marlatt, G. A. (2011). *Mindfulness-based relapse prevention for addictive behaviors.* Guilford Press.
• Bowen, S., & Marlatt, G. A. (2009). Surfing the urge. *Psychology of Addictive Behaviors*, 23(4), 666-671.
• Bowen, S., Witkiewitz, K., et al. (2014). Relative efficacy of MBRP. *JAMA Psychiatry*, 71(5), 547-556.
• Brewer, J. A., et al. (2011). Mindfulness training for smoking cessation. *Drug and Alcohol Dependence*, 119(1-2), 72-80.
• Chiesa, A., & Serretti, A. (2014). Are MBIs effective for SUDs? *Substance Use & Misuse*, 49(5), 492-512.
• Franken, I. H. A., et al. (2002). Initial validation of two opiate craving questionnaires. *Addictive Behaviors*, 27(5), 675-685.

41 References (2/4)

- Garland, E. L., Froeliger, B., & Howard, M. O. (2014). Mindfulness targets neurocognitive mechanisms. *Frontiers in Psychiatry*, 4, 173.
- Garland, E. L., et al. (2016). MORE vs. CBT for co-occurring SUDs. *J. Consulting and Clinical Psychology*, 84(4), 281-293.
- Ghosh, A., Basu, D., & Avasthi, A. (2018). Relapse in opioid dependence: Indian perspective. *Indian J Psychiatry*, 60(S4), S469-S476.
- Glasner-Edwards, S., et al. (2017). MBRP for stimulant dependent adults. *Mindfulness*, 8(1), 126-135.
- Grant, S., et al. (2017). MBRP for SUDs: Meta-analysis. *J. Addiction Medicine*, 11(5), 386-396.
- Humeniuk, R., et al. (2008). Validation of ASSIST. *Addiction*, 103(6), 1039-1047.
- Jain, R., Majumder, P., & Gupta, T. (2013). *Indian Journal of Psychiatry*, 55(Suppl 1), S86-S92.
- Kabat-Zinn, J. (1990). *Full catastrophe living.* Delacorte Press.

42 References (3/4)

- Karyadi, K. A., et al. (2014). Trait mindfulness and substance use. *Drug and Alcohol Dependence*, 143, 1-10.
- Li, W., et al. (2017). Mindfulness for substance misuse: Meta-analysis. *J. Substance Abuse Treatment*, 75, 62-96.
- Marlatt, G. A., & Gordon, J. R. (1985). *Relapse prevention.* Guilford Press.
- Mattoo, S. K., et al. (2009). Psychosocial factors in relapse. *Indian J. Medical Research*, 130(6), 702-708.
- Ministry of Social Justice (2019). *Magnitude of substance use in India.* Government of India.
- Moeller, F. G., et al. (2001). Psychiatric aspects of impulsivity. *Am. J. Psychiatry*, 158(11), 1783-1793.
- Murphy, C., & MacKillop, J. (2012). Impulsivity, mindfulness, and alcohol. *Psychopharmacology*, 219(2), 527-536.
- NIDA. (2020). *Drugs, brains, and behavior: The science of addiction.*

43 References (4/4)

- Patton, J. H., Stanford, M. S., & Barratt, E. S. (1995). Factor structure of BIS. *J. Clinical Psychology*, 51(6), 768-774.
- Robinson, T. E., & Berridge, K. C. (1993). Neural basis of drug craving. *Brain Research Reviews*, 18(3), 247-291.
- Sarkar, S., & Balhara, Y. P. S. (2016). *Indian Journal of Psychiatry*, 58(3), 290-295.
- Serre, F., et al. (2015). Ecological momentary assessment in craving. *Addiction*, 110(7), 1070-1082.
- Stanford, M. S., et al. (2009). Fifty years of the BIS. *Personality and Individual Differences*, 47(5), 385-395.
- Tabachnick, B. G., & Fidell, L. S. (2013). *Using multivariate statistics* (6th ed.). Pearson.
- Tiffany, S. T., & Wray, J. M. (2012). Clinical significance of drug craving. *Ann NY Acad Sci*, 1248(1), 1-17.
- WHO ASSIST Working Group. (2002). ASSIST development. *Addiction*, 97(9), 1183-1194.
- Witkiewitz, K., et al. (2013). MBRP for craving. *Addictive Behaviors*, 38(2), 1563-1571.

Thank You

Questions & Discussion

Tejas Dangodra

2nd Year MPhil Trainee in Clinical Psychology

Man College of Special Education & Psychological Studies

Krantivir Tatya Tope Vishwavidyalaya, Guna, MP

tejasdangodra99@gmail.com | +91 8140171722